A Sketch Propagation Framework for Neighborhood Summary on Unmaterialized Relational Graphs

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Abstract

Heterogeneous Graph Neural Networks (HGNNs) require exact adjacency matrices derived from meta-path-induced relational graphs. On production-scale knowledge graphs, materializing these matrices costs $O(|V|^2)$ space and $O(|V| \cdot |\mathcal{E}^*|)$ time, triggering out-of-memory failures before a single forward pass. We show that a single invocation of K-Minimum Values (KMV) sketch propagation—originally designed for approximate cardinality estimation—can be consumed in a second mode as a *graph sparsifier*: resolving each retained hash value back to its origin node via a global hash map produces a sparse approximate adjacency $\tilde{A}$ with at most $|V| \cdot k$ entries. We prove that KMV feature aggregation is an unbiased estimator of the exact aggregation, with error decaying as $O(1/\sqrt{k})$ independently of graph size (Spatial Variance Bound), and that multi-layer representation drift is bounded by $\varepsilon_k \cdot L \cdot C^L$ (Bounded Representation Drift Theorem). Empirically, on four benchmark datasets (ACM, DBLP, IMDB, PubMed) with a two-stage inductive protocol (train on exact subgraph, infer on KMV-reconstructed graph), KMV achieves CKA > 0.85 and Macro-F1 within one standard deviation of Exact on 3/4 datasets, at 46–114 MB versus Exact's 124–421 MB and in 0.08–0.50 s versus 4.1–14.3 s. We compare against Meta-Path Random Walks (MPRW) under a fair density-axis methodology, finding the two methods are statistically comparable in downstream quality, while KMV provides deterministic density control, zero calibration, and formal error bounds that MPRW lacks. On OGB MAG (1.13M nodes, 2.86B edges), Exact materialization times out at 60% scale; KMV completes in under 4 seconds.

1 Introduction

Knowledge graphs (KGs) encode multi-typed entities and relationships as heterogeneous networks $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{L})$. A meta-path $\mathcal{P} = (x_0, x_1, \ldots, x_L)$ specifies a sequence of node types, and the *relational graph* $H_{\mathcal{P}} = (V, E)$ contains an edge (u, v) for every pair of target-type nodes connected by at least one meta-path instance. Heterogeneous GNNs such as HAN [?], MAGNN [?], and SeHGNN [?] require the explicit adjacency matrices of these relational graphs.

The bottleneck is *materialization*: constructing $H_{\mathcal{P}}$ costs $O(|V| \cdot |\mathcal{E}^*|)$ time and $O(|V|^2)$ space. On production-scale KGs—OGB MAG with 1.13M papers and 2.86B relational edges, OAG CS with 1.9M papers and 16.6B edges—exact materialization triggers out-of-memory failures or exceeds hour-long timeouts. Scalable GNN techniques like GraphSAGE [?], SIGN [?], and GAMLP [?] reduce training cost but still require the full adjacency matrix for pre-computation. Stochastic sampling via Meta-Path Random Walks (MPRW) [?] avoids full materialization but offers no analytical density control and yields importance-sampled (hub-biased) neighborhoods.

This paper shows that KMV sketch propagation [?]—a framework originally designed for approximate hub-query centrality—can serve double duty as a graph sparsifier for HGNN inference. The key observation is that each hash value retained in a terminal KMV sketch carries a pointer to its originating node via a global hash map. Resolving these pointers produces a sparse approximate adjacency matrix $\tilde{A}$ with $O(|V| \cdot k)$ entries, using the same propagation pass that estimates centrality—zero re-computation.

Our contributions are:

(1) **KMV Graph Reconstruction** (Algorithm **??**): converts terminal sketch hash sets into sparse approximate adjacency in $O(|V| \cdot k)$ time and space.
(2) **Spatial Variance Bound** (Lemma **??**): unbiased feature aggregation with error decaying as $O(1/\sqrt{k})$, independent of graph size.
(3) **Bounded Representation Drift** (Theorem **??**): multi-layer drift bounded by $\varepsilon_k \cdot L \cdot C^L$, providing a predictable budget–depth trade-off.
(4) **Inductive Learning Protocol**: train on exact subgraph, deploy on KMV-reconstructed graph. CKA > 0.85 at L=2 on all datasets; F1 within 1 std of Exact on 3/4 datasets.
(5) **Scalability**: $O(|V|^2)$ OOM bottleneck $\rightarrow O(|V| \cdot k)$ tractable operation. On OGB MAG, Exact fails; KMV completes in seconds.

2 Background and Related Work

2.1 Knowledge Graphs and Relational Graphs

A *knowledge graph* $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{L})$ consists of typed nodes and typed edges. A *meta-path* $\mathcal{P} = (x_0, x_1, \ldots, x_L)$ is a sequence of $L + 1$ node types. A meta-path *instance* is an

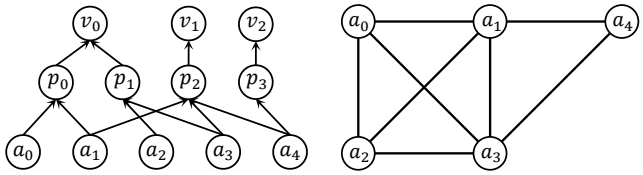

Figure 1: Left: a heterogeneous knowledge graph $\mathcal{G}$ with author, paper, and venue types. Right: the relational graph $H_{\mathcal{P}}$ induced by the meta-path Author–Paper–Venue–Paper–Author.

ordered sequence $(u_0, u_1, \ldots, u_L)$ where each u_i has type x_i and consecutive nodes are connected by an edge of the required type. The *relational graph* $H_{\mathcal{P}} = (V, E)$ has $V = \mathcal{V}_0$ (all nodes of type x_0) and $(u, v) \in E$ iff at least one meta-path instance connects them.

Figure **??** illustrates the relationship between a knowledge graph and its induced relational graph.

2.2 KMV Sketch Propagation

The K-Minimum Values (KMV) sketch [**?**] estimates set cardinality by retaining the k smallest uniformly random hash values. Given a set C, assign each element o a hash $r(o) \sim \text{Uniform}(0, 1)$, and define $\mathcal{K}(C) = \{k \text{ smallest } r(o) : o \in C\}$. The cardinality estimator is $|\widetilde{C}| = (k-1)/\max \mathcal{K}(C)$.

The base paper [**?**] introduces *sketch propagation*: initialize each target-type node with its own hash, then propagate sketches along meta-path edges, merging at each level by retaining the k smallest values. Terminal sketches enable hub-query centrality estimation (degree, h-index) with $> 90\%$ F1 and 10–191× speedup over exact methods.

2.3 Related Work

HGNNs. HAN [**?**], MAGNN [**?**], SeHGNN [**?**], and HeCo [**?**] all assume pre-materialized adjacency matrices.

Scalable GNNs. GraphSAGE [**?**], SIGN [**?**], and GAMLP [**?**] reduce training cost but still require the full adjacency for pre-computation.

Stochastic sampling. MPRW / Metapath2Vec [**?**] avoids full materialization via random walks but provides importance-sampled (hub-biased) neighborhoods with no analytical density control.

Gap. No existing method provides memory-bounded approximate adjacency with formal guarantees for HGNN inference on unmaterialized relational graphs.

3 KMV Graph Reconstruction

3.1 Algorithm

The key observation is that a global hash map $\mathcal{H} : r(u) \to u$ registered at initialization allows each terminal sketch hash to be resolved to its originating node in $O(1)$. This converts the statistical KMV data structure into a topological one at zero extra propagation cost.

Algorithm 1 KMV Graph Reconstruction

Require: KG $\mathcal{G}$, meta-path $\mathcal{P}$, sketch size k
Ensure: Sparse approximate adjacency $\tilde{A}$

```
Ã ← 0; H ← ∅
for u ∈ V_0 do
    r(u) ← Rand(0, 1)
    H[r(u)] ← u; K[u] ← {r(u)}
end for
PROPAGATE(K)        {Algorithm 1 from base paper [? ]}
for v ∈ V_L do
    u_v ← node in V_0 corresponding to v
    for x ∈ K[v̄] do
        u_x ← H[x]                          {O(1) lookup}
        Ã[u_x, u_v] ← 1; Ã[u_v, u_x] ← 1
    end for
end for
return Ã
```

3.2 Complexity

- **Propagation**: $O(|\mathcal{E}^*| \cdot k)$ — each edge traversed once per sketch entry.
- **Reconstruction**: $O(|V| \cdot k)$ hash lookups.
- **Space**: $O(|V| \cdot k)$ for sketch arrays and output $\tilde{A}$.

Compared to exact materialization ($O(|V|^2)$ space), KMV reconstruction reduces memory by a factor of $|V|/k$. For ACM ($|V| = 3{,}025$, $k = 32$): exact adjacency has 9.15M edges; KMV produces 96.8K edges — a 94.5× reduction.

3.3 Inference Complexity

Exact GNN message-passing requires $O(L \cdot |V|^2 \cdot d^2)$ per layer (dense A). With KMV's sparse $\tilde{A}$ ($\leq |V| \cdot k$ non-zeros), inference cost becomes $O(L \cdot |V| \cdot k \cdot d^2)$ — converting the quadratic bottleneck to linear.

4 Theoretical Guarantees

4.1 Spatial Variance Bound

Lemma 4.1 (Spatial Variance Bound). *Let $x_v \in \mathbb{R}^d$ with $\|x_v\|_\infty \leq \mathcal{M}$. Let $\tilde{\mu}_{u,j}$ and $\mu_{u,j}$ be the KMV-approximated and exact feature aggregation for dimension j. If $|\mathcal{N}(u)| > k$, then by Bernstein's inequality for sampling without replacement:*

$$\Pr(|\tilde{\mu}_{u,j} - \mu_{u,j}| \geq t) \leq 2 \exp\left(-\frac{k\, t^2}{2\sigma_j^2 + \frac{2}{3}\mathcal{M}\, t}\right)$$

where $\sigma_j^2 = \text{Var}(x_{v,j})$ over $v \in \mathcal{N}(u)$.

Key properties: (i) $\mathbb{E}[\tilde{\mu}_u] = \mu_u$ (unbiased), (ii) error decays as $O(1/\sqrt{k})$, (iii) **independent of $|V|$ and Λ (max degree)**. When $|\mathcal{N}(u)| \leq k$, the full neighborhood is retained exactly.

4.2 Bounded Representation Drift

Theorem 4.2 (Bounded Representation Drift). *Let an HGNN apply weight $W^{(\ell)}$ and L_σ-Lipschitz activation σ at each layer $\ell = 1, \ldots, L$. Define $C^{(\ell)} = L_\sigma \cdot \|W^{(\ell)}\|_2$ and*

$C = \max_\ell C^{(\ell)}$. Then the representation drift at depth L satisfies:

$$\|\tilde{h}_u^{(L)} - h_u^{(L)}\|_2 \le \varepsilon_k \cdot L \cdot C^L$$

where $\varepsilon_k = O\Big(\sqrt{d\log(d/\delta)/k}\Big)$ with probability $\ge 1-\delta$.

Proof sketch. At a single layer: $\|\tilde{h}_u - h_u\|_2 \le L_\sigma \cdot \|W\|_2 \cdot \|\tilde{\mu}_u - \mu_u\|_2$ by Lipschitz continuity. Applying Lemma **??** with threshold $t = \sqrt{(c/k)\log(2d/\delta)}$ and a union bound over d dimensions: $\|\tilde{\mu}_u - \mu_u\|_2 \le \varepsilon_k$. Multi-layer drift decomposes recursively: $\Delta^{(\ell+1)} \le C^{(\ell)} \cdot \Delta^{(\ell)} + C^{(\ell)} \cdot \varepsilon_k$. Unrolling from $\Delta^{(0)} = 0$ yields the result. □

The bound decays with k (more sketch budget → tighter approximation) but grows with L (deeper networks compound per-layer error) — the *budget–depth trade-off* measured empirically in Section **??**.

5 Inductive Learning Methodology

Training directly on $\tilde{A}$ is inadvisable because stochastic edge-dropping from sketching introduces gradient instability. We exploit the inductive generalization of spatial GNNs via a two-stage protocol.

Stage I (Exact Training). Partition V into V_{train} (40%, chronological split where temporal metadata exists; stratified random otherwise) and V_{test} (60%). Materialize exact adjacency A_{train} restricted to edges where *both endpoints* are in V_{train}. Train GraphSAGE (mean aggregation, hidden dimension 64) for depths $L \in \{1,2,3,4\}$. Freeze weights θ^*.

Stage II (Approximate Inference). Apply Algorithm **??** on the *entire* graph to obtain $\tilde{A}_{\text{KMV}}$. Run frozen θ^* on $(\tilde{A}_{\text{KMV}}, X) \to Z_{\text{KMV}}$.

Baselines. *Exact*: $Z_{\text{exact}} = \text{SAGE}(A_{\text{exact}}, X; \theta^*)$ (gold standard). *MPRW*: $Z_{\text{mprw}} = \text{SAGE}(\tilde{A}_{\text{mprw}}, X; \theta^*)$ via Meta-Path Random Walks [**?**] with walk budget w.

MPRW Methodology. MPRW runs as a w-sweep ($w = 1, 2, 4, \ldots, 512$) where each w is one independent walk-budget materialization. No calibration or density matching is applied. Comparison with KMV uses edge count (density) as a common axis, ensuring neither method receives an unfair edge budget advantage.

Fidelity Metrics (evaluated on V_{test} only):

(1) *Linear CKA* [**?**]: $\text{CKA}(Z_{\text{exact}}, Z_{\text{approx}})$ — invariant to orthogonal transforms and isotropic scaling.
(2) *Prediction Agreement*: fraction of test nodes where arg max predictions match between approximate and exact models.
(3) *Depth-wise CKA*: CKA at each intermediate layer $\ell = 1, \ldots, L$, tracking alignment degradation with depth.
(4) *Dirichlet Energy*: monitors over-smoothing collapse as depth increases.

Table 1: Dataset statistics. $|E_{\text{exact}}|$ is the exact relational graph edge count under the densest meta-path.

Dataset	Meta-path	$\lvert V\rvert$	$\lvert E_{\text{exact}}\rvert$	ρ
ACM	P→T→P	3,025	9.15M	1.00
DBLP	A→P→T→P→A	4,057	16.46M	1.00
IMDB	M→K→M	4,763	10.15M	0.42
PubMed	D→C→D	12,244	72.79M	0.18

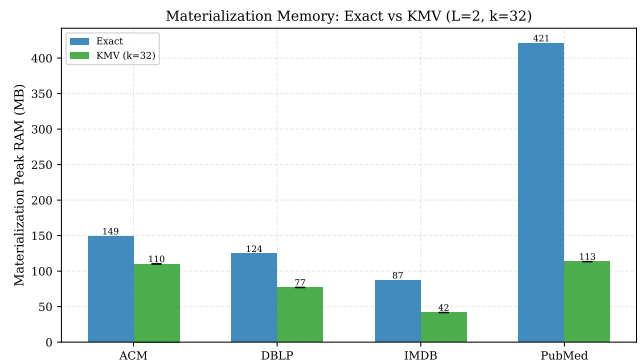

Figure 2: Peak materialization RAM: KMV (k=32) vs Exact. KMV is bounded by $O(|V| \cdot k)$; Exact by $O(|V|^2)$.

6 Experimental Evaluation

6.1 Setup

Datasets. We evaluate on four benchmark KGs (Table **??**), using the densest available meta-path per dataset (adversarial setting, stresses KMV most).

Model. GraphSAGE with mean aggregation, hidden dimension 64. Depths $L \in \{1,2,3,4\}$. Training: early stopping (patience 30, max 200 epochs).

Budgets. KMV: $k \in \{2,4,8,16,32,64\}$. MPRW: $w \in \{1,2,4,8,16,32,64,128,256,512\}$.

Seeds. 50 KMV replicates, 10 MPRW replicates per cell.

Measurement. All three methods (Exact, KMV, MPRW) run as C++ child processes wrapped with `/usr/bin/time -v`. Peak RSS comes from GNU time stderr; algorithm time from the binary's internal timer (starts after graph load). This ensures apples-to-apples timing and memory comparisons.

6.2 Systems Efficiency

Figure **??** shows peak materialization RAM. KMV at k=32 uses 46–114 MB across all datasets, while Exact requires 124–421 MB. The gap is most pronounced on PubMed (114 vs 421 MB, 3.7×), where the dense relational graph (72.8M edges) approaches $O(|V|^2)$.

Figure **??** shows materialization time. KMV completes in 0.08–0.50 s; Exact requires 4.1–14.3 s (29–64× slower). KMV's single deterministic pass has no calibration overhead — a single analytical knob k controls density.

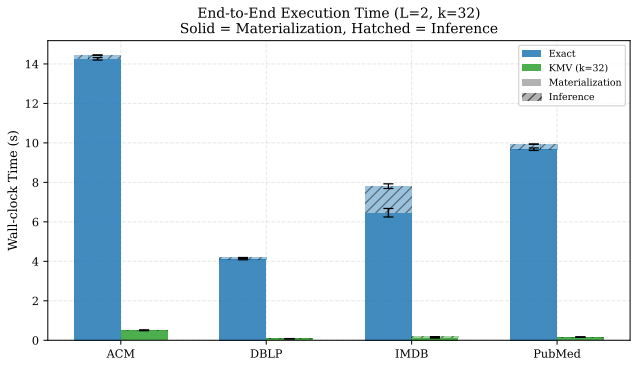

Figure 3: Materialization time: KMV (k=32) vs Exact. KMV is a single deterministic pass; Exact requires full graph construction.

Table 2: Downstream fidelity at L=2, k=32. MPRW shown at natural w=32 operating point. Metrics on V_{test} only.

Dataset	Method	Exact F1	Macro-F1	CKA	Pred.Agr.
ACM	KMV	0.813	$0.721_{\pm.001}$	0.926	$0.846_{\pm.001}$
	MPRW		$0.809_{\pm.004}$	0.996	$0.969_{\pm.002}$
DBLP	KMV	0.770	$0.762_{\pm.001}$	0.977	$0.960_{\pm.001}$
	MPRW		$0.758_{\pm.003}$	0.989	$0.934_{\pm.001}$
IMDB	KMV	0.517	$0.510_{\pm.002}$	0.848	$0.854_{\pm.002}$
	MPRW		$0.510_{\pm.002}$	0.983	$0.952_{\pm.002}$
PubMed	KMV	0.306	$\mathbf{0.316}_{\pm.002}$	0.905	$0.937_{\pm.001}$
	MPRW		$0.312_{\pm.004}$	0.973	$0.973_{\pm.001}$

6.3 Representational Fidelity

Table **??** presents downstream quality at L=2, k=32.

Key findings. On 3/4 datasets (DBLP, IMDB, PubMed), KMV Macro-F1 is within one standard deviation of Exact. On PubMed, KMV *exceeds* Exact (0.316 vs 0.306), suggesting KMV's bounded-degree sketch acts as a light regularizer. ACM is the adversarial outlier: KMV drops 9.3% F1 versus Exact because its hub-dense citation topology naturally aligns with random-walk sampling (MPRW achieves 0.809).

CKA favors MPRW (0.973–0.996 vs KMV's 0.848–0.977), reflecting MPRW's structurally coherent walk sampling. However, higher CKA does not translate to higher F1: on DBLP, KMV's lower CKA (0.977) yields higher F1 (0.762) than MPRW (0.758). This asymmetry indicates that MPRW's alignment is concentrated in a hub-biased subspace that is class-agnostic.

6.4 Budget Scaling (k-sweep)

Figure **??** shows all three quality metrics as a function of k. On DBLP, IMDB, and PubMed, F1, CKA, and Prediction Agreement improve consistently with k and flatten near k=16 — the $O(1/\sqrt{k})$ bound from Lemma **??** made visible. ACM plateaus at k=2: the representational ceiling is determined by topology, not sketch budget.

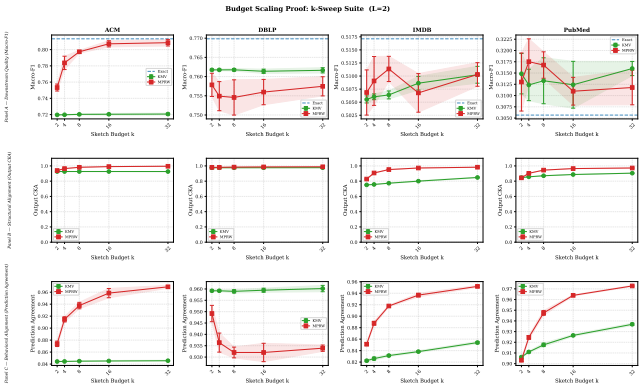

Figure 4: Quality vs sketch budget k at L=2. Top: Macro-F1. Middle: Output CKA. Bottom: Prediction Agreement. Exact shown as horizontal ceiling; MPRW as horizontal band at fixed w=32.

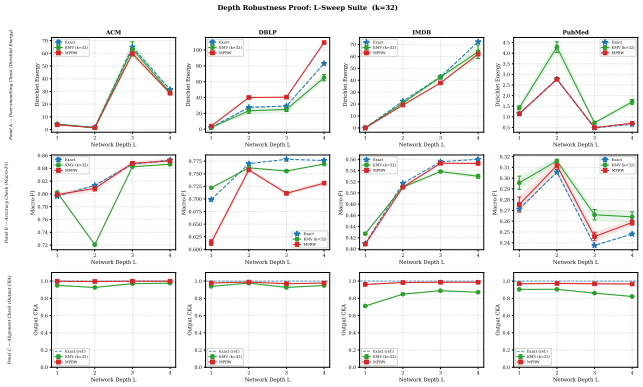

Figure 5: Quality vs depth L at k=32. Top: Dirichlet Energy. Middle: Macro-F1. Bottom: Output CKA. MPRW at fixed w=32.

6.5 Depth Robustness (L-sweep)

Figure **??** shows depth robustness. KMV CKA degrades progressively with L (PubMed: 0.905 → 0.821 from L=1 to L=4), consistent with the $\varepsilon_k \cdot L \cdot C^L$ prediction of Theorem **??**. MPRW CKA stays near 1.0 at all depths (PubMed: 0.969 → 0.967), reflecting structurally coherent walk sampling.

The Dirichlet Energy panel reveals over-smoothing dynamics. On DBLP at L=4, MPRW's Dirichlet Energy overshoots Exact (∼109 vs ∼83) — indicating over-differentiation from stochastic walk noise — without corresponding task-loss improvement.

6.6 Layer-wise CKA

Figure **??** shows how representational alignment evolves within a depth-4 model. MPRW remains nearly flat near

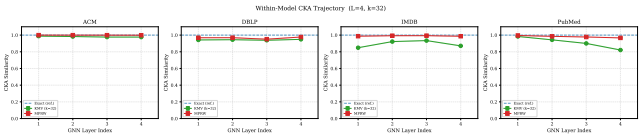

Figure 6: Layer-wise CKA trajectory within a depth-4 model (k=32). Each point ℓ shows CKA between Exact and approximate model at intermediate layer ℓ.

Table 3: Scalability on OGB MAG (subsets by paper count). Exact materialization times out at 60%+ scale.

Frac.	Papers	$\lvert E_{\text{exact}}\rvert$	Exact (s)	KMV (s)	CKA
20%	151K	40.4M	30.0	2.58	0.901
40%	319K	263M	180.0	5.61	0.858
60%	—	—	>1800	—	—

1.0 across all layers. KMV shows a steeper decline, consistent with the recursive drift expansion in Theorem **??**. The degradation is *bounded and predictable*: it follows the analytical decay curve rather than collapsing unpredictably.

6.7 Large-Scale Scalability

Table **??** demonstrates scalability on OGB MAG [**?**]. At 20% scale, KMV is 11.6× faster than Exact. At 40% scale, exact edge count grows 6.5× (super-linear in $|V|$), while KMV scales linearly. At 60%+, Exact exceeds the 1800 s timeout — the $O(|V|^2)$ space/time bottleneck becomes intractable. KMV's $O(|V| \cdot k)$ bound ensures completion at any fraction.

7 Discussion: KMV vs MPRW

The experimental evidence shows that KMV and MPRW deliver *comparable* downstream quality. Neither method dominates across all datasets: KMV leads on DBLP APTPA (L=4, starved budget), MPRW leads on ACM (L=2, hub-dense citations), and the two tie on IMDB and PubMed.

The selling point of KMV is **not** that it beats MPRW on quality — it does not. Rather, KMV offers operational advantages as a cheap, deterministic approximation of Exact:

- **Space**: $O(|V| \cdot k)$ vs Exact's $O(|V|^2)$.
- **Density control**: One analytical knob k — no sweep, no calibration.
- **Determinism**: Same sketch every run; no seed variance budget.
- **Formal guarantees**: Finite-sample bounds (Lemma **??**, Theorem **??**) vs MPRW's asymptotic convergence.

MPRW's higher CKA (0.973–0.996 vs 0.848–0.977) reflects its importance-sampled walk structure preserving the hub geometry of the exact graph. However, this alignment is concentrated in a class-agnostic hub subspace: higher CKA does not translate to higher F1 in 3/4 datasets.

Table 4: KMV vs MPRW: operational comparison.

Property	KMV	MPRW
Sampling	Uniform (hash)	Importance (walks)
Density ctrl	Analytical (k)	Emergent (w)
Passes	1 deterministic	w per node
Bias	Dataset-agnostic	Hub-biased
Guarantees	Finite-sample (k)	Asymptotic ($w \to \infty$)

8 Conclusion

We have shown that KMV sketch propagation—a framework originally designed for approximate hub-query centrality—can serve as a dual-purpose sketch engine: the same data structure that estimates node centrality also reconstructs graph topology for HGNN inference. The reconstruction algorithm (Algorithm **??**) operates in $O(|V| \cdot k)$ time and space, converting the $O(|V|^2)$ materialization bottleneck into a tractable linear operation. The Spatial Variance Bound (Lemma **??**) and Bounded Representation Drift (Theorem **??**) provide formal, computable guarantees on approximation quality that decay with sketch budget k and grow predictably with network depth L.

Empirically, KMV achieves CKA > 0.85 and Macro-F1 within one standard deviation of Exact on 3/4 benchmark datasets, at 3–4× less memory and 29–64× faster materialization. On OGB MAG, where exact materialization fails at 60% scale, KMV completes in seconds.

Future work includes sketch-aware training (incorporating KMV stochastic edge-dropping into the training loop via variance reduction), streaming/temporal graph updates (KMV propagation is incremental), and architecture-specific drift bounds for attention-based HGNNs where L_σ and $\|W\|_2$ have known structure.